

13.2 Math and the Environment



Figure 13.8 Solar panels harness the sun's energy to power homes, businesses, and various methods of transportation. (credit: modification of work “Craters of the Moon solar array” by NPS Climate Change Response/Flickr, Public Domain Mark 1.0)

Learning Objectives

After completing this section, you should be able to:

1. Compute how conserving water can positively impact climate change.
2. Discuss the history of solar energy.
3. Compute power needs for common devices in a home.
4. Explore advantages of solar power as it applies to home use.

Climate change and emissions management have been debated topics in recent years. However, more and more people are recognizing the impacts that have resulted in temperature changes and are seeking timely and effective action. The World Meteorological Organization shared in a June 2021 publication that “2021 is a make-or-break year for climate action, with the window to prevent the worst impacts of climate change—which include ever more frequent more intense droughts, floods and storms—closing rapidly.” The problem no longer belongs to a few countries or regions but rather is a worldwide concern measured with increasing temperatures leading to decreased glacier coverage and resulting rise in sea levels.

The good news is, there are small steps that each of us can do that collectively can positively impact climate change.

Making a Positive Impact on Climate Change—Water Usage

Our use of water is one element that impacts climate change. Having access to clean, potable water is critical for not only our health but also for the health of our ecosystem. About 1 out of 10 people on our planet do not have easy access to clean water to drink. As each of us conserves water, we prolong the life span of fresh water from our lakes and rivers and also reduce the impact on sewer systems and drainage in our communities. Additionally, as we conserve water, we also conserve electricity that is used to bring water to and in our homes. So, what can we do to help conserve water?

EXAMPLE 13.5

Brushing Your Teeth (One Person's Contribution)

Brushing your teeth with the water running continually uses about 4 gal of water. Turning the faucet off when you are not rinsing uses less than one-fourth of a gallon of water. Considering the recommendation to brush your teeth twice a day, how much water would be saved in a week if the faucet was off when not rinsing?

Solution

Leaving the water running continually:

Step 1: Calculate gallons used not with water running continually:

Brushing twice a day for 7 days using 4 gal of water for each brushing

$$(2 \text{ times a day})(7 \text{ days})(4 \text{ gal}) = 56 \text{ gal}$$

Step 2: Calculate gallons used turning the faucet off when you are not rinsing:

Brushing twice a day for 7 days using 0.25 gal of water for each brushing

$$(2 \text{ times a day})(7 \text{ days})(0.25 \text{ gal}) = 3.5 \text{ gal}$$

Step 3: Calculate savings:

$$\begin{aligned} \text{Savings} &= 56 \text{ gal} - 3.5 \text{ gal} \\ &= 52.5 \text{ gal} \end{aligned}$$

During one week, 52.5 gal of water would be saved if one person turned the faucet off except when rinsing when brushing your teeth.

YOUR TURN 13.5

1. Brushing your teeth with the water running continually uses about 4 gal of water. Turning the faucet off when you are not rinsing uses less than one-fourth of a gallon of water. Consider the recommendation to brush your teeth twice a day, how much water would be saved in a month (4 weeks) if the faucet was off when not rinsing?

EXAMPLE 13.6

Brushing Your Teeth (Multiple People's Contribution – Town)

Using the data in [Example 13.5](#), how much water would be saved in a month if one-fifth of a town's population of 15,000 turned the faucet off when brushing their teeth except when rinsing?

Solution

Step 1: From [Example 13.5](#), we found that 1 person saves 52.5 gal per week.

Step 2: Calculate the population to save water:

$$\text{One-fifth of 15,000 people} = 3,000 \text{ people}$$

Step 3: One-fifth of a town's population turning off the faucet when brushing their teeth for a month:

$$(3,000)(52.5 \text{ gal per week})(4 \text{ weeks}) = 630,000 \text{ gal}$$

During one month, 630,000 gallons of water would be saved if one-fifth of a town of 15,000 people turned the faucet off except when rinsing when brushing their teeth.

YOUR TURN 13.6

1. Using the data in Your Turn 13.5, how much water would be saved in a month if one-third of a town's population of 15,000 turned the faucet off when brushing their teeth except when rinsing?

EXAMPLE 13.7

Brushing Your Teeth (Multiple People's Contribution – State)

Using the data in [Example 13.5](#), how much water would be saved in a year if one-fourth of the population of the state of Minnesota, which is approximately 5.6 million people, turned the faucet off when brushing their teeth except when rinsing for a year (52 weeks)?

Solution

Step 1: From [Example 13.5](#), we found that 1 person saves 52.5 gal per week.

Step 2: Calculate the population to save water:

$$\text{One-fourth of 5.6 million people} = 1.4 \text{ million people}$$

Step 3: One-fourth of a town's population turning off the faucet when brushing their teeth for a month:

$$(1.4 \text{ million})(52.5 \text{ gal per week})(52 \text{ weeks}) = 3,822 \text{ million gal}$$

During one year, 3,822 million gal of water would be saved if one-fourth of the state of Minnesota turned the faucet off except when rinsing when brushing their teeth.

> YOUR TURN 13.7

1. Using the data in Example 13.5, how much water would be saved in a year if one-sixth of the population of the state of Florida, which is approximately 21.6 million people, turned the faucet off when brushing their teeth except when rinsing for a year (52 weeks)?

History of Solar Energy

In the mid-1800s, Willoughby Smith discovered photoconductive responsiveness in selenium. Shortly thereafter, William Grylls Adams and Richard Evans Day discovered that selenium can produce electricity if exposed to the sun was a major breakthrough. Less than 10 years later, Charles Fritts invented the first solar cells using selenium. Jumping a mere 100 years later, Bell Labs in the United States produced the first practical photovoltaic cells in the mid-1950s and developed versions used to power satellites in the same decade.

Solar panel use has exploded in recent decades and is now used by residences, organizations, businesses, and government buildings such as the White House, space to power satellites, and various methods of transportation. One reason for the expansion is a continuing drop in cost combined with an increase in performance and durability. In the mid-1950s, the cost of a solar panel was around \$300 per watt capability. Twenty years later, the cost was a third of the 1950s' cost. Currently, solar panel cost has dropped to less than \$1 per watt while decreasing in size as well as increasing in longevity. The dropping price and improved performance has moved solar to a modest investment that can pay for itself in less than half the time of systems from 15 years ago.

? WHO KNEW?

Solar Power's Age

The sun has been harnessed by humans for centuries. The earliest recorded use of tapping the sun's energy for power dates back to the seventh century BC when man focused the sun's rays through a magnifying glass to create fire. Four thousand years later, we find historical record of using mirrors to focus the sun and light torches, often for ceremonial proceedings. Use of the sun to light torches continued through the centuries and has been recorded by various cultures including the Chinese civilization in 20 AD and beyond.

In more recent years, the sun was harnessed to power ovens on ships traversing oceans in the 1700s. At the same time, the power of the sun was utilized to power steamboats through the 1800s. Mária Telkes, a Hungarian-born American scientist, invented a widely deployed solar seawater distiller used on World War II life rafts. Soon after, she partnered with architect Eleanor Raymond to design the first modern home to be completely heated by solar power. Air warmed on rooftop collectors transferred heat to salts, which stored the heat for later use.

Although solar panels as we know them today are relatively new in history, use of the sun to harness power is much older.

Compute Power Needs for Common Home Devices

A **kilowatt** (kW) is 1,000 watts (W). A **kilowatt-hour** (kWh) is a measurement of energy use, which is the amount of energy used by a 1,000-watt device to run for an hour. Using the definition of a kilowatt-hour, to calculate how long it would take to consume 1 kWh of power, we divide 1,000 by the watts use of a device.

FORMULA

$$1,000/\text{watts} = \text{time needed to use 1 kW}$$

For example, a 75 W bulb would take $1,000 \div 75 = 13.3$ hours to use 1 kW of power.

FORMULA

$$\text{watts}/1,000 = \text{kilowatt hours}$$

EXAMPLE 13.8**Calculating the Kilowatt-Hours Needed to Run a Television**

A 48 in plasma television uses about 200 W. How many kilowatt-hours are needed to run the television in a month if the television is on for an average of 2.5 hours a day?

✓ **Solution**

Step 1: $1,000/(200 \text{ watts}) = 5 \text{ hours to use 1 kW}$

Step 2: $(2.5 \text{ hours a day})(30 \text{ days}) = 75 \text{ hours of use}$

Step 3: $75 \text{ hour}/5 \text{ hours per kW} = 15 \text{ kW}$

The television will consume about 15 kW in a month.

**YOUR TURN 13.8**

1. A 40 in plasma television uses about 175 W. How many kilowatt-hours are needed to run the television in a month if the television is on for an average of 3 hours a day?

EXAMPLE 13.9**Calculating the Cost to Run a Refrigerator**

A medium-sized Energy Star-rated refrigerator uses about 575 W and runs for about 8 hours per day. What is the monthly (30 days) cost of running the refrigerator if the electric rate is 12 cents per kilowatt-hour?

✓ **Solution**

Step 1: Calculate the watts per day:

$$(575 \text{ W})(8 \text{ hours}) = 4,600 \text{ W per day}$$

Step 2: Calculate the kilowatt-hours.

$$(4,600)/(1,000) = 4.6 \text{ kWh}$$

Step 3: Calculate the daily cost.

$$(4.6 \text{ kWh})(12 \text{ cents}) = 55 \text{ cents} = \$0.55$$

Step 4: Calculate the monthly cost.

$$(\$0.55)(30) = \$16.50$$

It would cost about \$16.50 to run the refrigerator for a month.

**YOUR TURN 13.9**

1. A dorm-sized Energy Star-rated refrigerator uses about 375 W and runs for about 9 hours per day. What is the monthly cost of running the refrigerator if the electric rate is 14 cents per kilowatt-hour?

EXAMPLE 13.10**Calculating the Kilowatt-Hours to Run an Oven**

An electric oven is labeled as 4,000 W. How much would it cost to bake a cake for 30 minutes if the electric rate is 14 cents per kilowatt-hour?

 Solution

Step 1: Determine the time it takes to use 1 kW of power:

$$1,000/(4,000 \text{ watts}) = 0.25 \text{ hours to use 1 kW}$$

For every 15 minutes, the oven uses 1 kW of power.

Step 2: Determine how many kilowatt-hours are needed to bake the cake for 30 minutes:

$$(30 \text{ minutes})/(15 \text{ minutes per kW}) = 2 \text{ kW}$$

Step 3: Calculate the cost of the oven usage:

$$(2 \text{ kW})(14 \text{ cents per kWh}) = 28 \text{ cents}$$

It would cost about 28 cents to bake the cake.

 YOUR TURN 13.10

1. A toaster oven is labeled as 2,000 W. How much would it cost to warm leftovers from a meal for 15 minutes if the electric rate is 12 cents per kilowatt-hour?

Solar Advantages

There are multiple advantages that solar power can offer us today including reducing greenhouse gas and CO₂ emissions, powering vehicles, reducing water pollution, reducing strain on limited supply of other power options such as fossil fuels. We will look further at reducing greenhouse gas and CO₂ emissions.

Any gas that prevents infrared radiation from escaping Earth's atmosphere is a greenhouse gas. There are 24 currently identified greenhouse gases of which carbon dioxide is one. When measuring the impact of any of the greenhouse gases, the measurements are given in units of carbon dioxide emissions. For this reason, greenhouse gas and carbon dioxide have become interchangeable in discussions.

**PEOPLE IN MATHEMATICS**

Charles Fritts and Mohammad M. Atalla

Charles Fritts, a New York inventor, is credited with creating the first solar cell, which he installed on a rooftop in New York City in 1884. While the solar cell was not very efficient, having a rate of conversion between 1 to 2%, this was a major step early in solar power energy. Today's solar cells have an efficiency on average of 15 to 20%, which yields a notably higher impact. Nonetheless, the work that Fritts successfully completed marked the start of solar energy through the use of photovoltaic solar panels in the United States.

Mohamed M. Atalla was an Egyptian-born scientist who moved to the United States to complete his studies, and undertook research and development at Bell Laboratories in New Jersey. Many of the early efficiency gains in solar cells were due to his development of processes for using silicon within electronic devices. Atalla's work led to the invention of silicon transistors and microchips (including his own invention of the MOSFET, the most widely used transistor in the world), and quickly increased the efficiency of solar cells.

EXAMPLE 13.11**Calculating the Solar Power for Average Home Use in Kilowatts**

If a home uses approximately 30 kW of electricity per day, what size solar system would be needed to fuel 80% of a home's needs for a month (30 days)?

 Solution

$$(30 \text{ kW hours})(30 \text{ days})(0.80) = 720 \text{ kW}$$

A solar system capable of producing 720 kW a month would be needed.

 YOUR TURN 13.11

1. A tiny home uses approximately 12 kW of electricity per day. What size solar system would be needed to fuel 80% of a home's needs for a month (30 days)?

Check Your Understanding

4. What is the relationship between 1 kW and watts?
5. An average bath uses about 35 gal of water. A water-saving showerhead uses approximately 2 gal of water per minute. If a person typically takes one bath a week, how much water is saved by replacing the baths with showers lasting 5 minutes over the course of a month?
6. How long does it take for a 30 W Google Nest audio to use 1 kW of power?
7. How long would it take for a 60 W bulb to consume 1 kW of power? Round your answer to the nearest hour.

**SECTION 13.2 EXERCISES**

1. A typical showerhead uses 5 gal of water per minute. A water-saving showerhead uses approximately 2 gal of water per minute. How much water would one person save in a month if they take a 6-minute shower 4 times a week?
2. An average toilet uses 5 gal of water per flush. A high-efficiency toilet uses about 1.25 gal per flush. How much water would a household save in a week if the toilet was flushed 8 times a day?
3. When washing dishes, leaving the faucet running utilizes about 15 gal every 5 minutes, where filling the sink and turning the faucet off except to rinse the dishes uses about 5 gal for washing a dishpan-sized load of dishes. How much water is saved by not leaving the faucet running if it takes 10 minutes to wash a dishpan-sized load of dishes each day for a month containing 30 days?
4. Leaving the water running when washing your hands consumes about 4 gal of water, whereas turning the water off when lathering reduces the water used to 1 gal. How much water is saved in an apartment of 6 people in an academic term of 16 weeks if each person averages washing their hands 3 times a day?
5. How long would it take for an energy-saving light bulb to consume 1 kW of power if the bulb is rated at 7.5 W? Round your answer to the nearest hour.
6. How long would it take for a 120 W light bulb to consume 1 kW of power? Round your answer to the nearest hour.
7. A portable television uses about 80 W per hour. How many kilowatt-hours are needed to run the television during a 3-day trip if the television is run for an average of 5.5 hours a day?
8. A flat iron to straighten hair is rated at 331 W. If it is used for 15 minutes a day, 5 times a week, how much would it cost a user over the course of a month if the electric rate is 13 cents per kilowatt-hour? Round your answer to the nearest cent.
9. You purchase a window air conditioner for your apartment living room rated at 1,000 W. If you run the air conditioner for an average of 3 hours a day for a week, how much would it cost if the electric rate is 12 cents per kilowatt-hour?
10. A cabin uses approximately 25 kW of electricity per day. What size solar system would be needed to fuel 90% of the